

DroneWQ: A Python package for processing MicaSense multispectral drone imagery for aquatic remote sensing

Anna E. Windle^{1,2*}, Patrick C. Gray^{3*}, Alejandro Román⁴, Sergio Heredia⁴, Gabriel Navarro⁴, Temuulen Enkhtamir⁵, Kurtis Kwan⁵, Nidhi Khiantani⁵, and Greg M. Silsbe⁶

¹ NASA Goddard Space Flight Center, Greenbelt, MD, United States ² Science Systems and Applications, Inc., Lanham, MD, United States ³ School of Marine Sciences, University of Maine, Orono, ME, United States ⁴ Department of Ecology and Coastal Management, Institute of Marine Sciences of Andalusia (ICMAN-CSIC), Spanish National Research Council (CSIC), 11519 Puerto Real, Spain ⁵ Duke University, Durham, NC, United States ⁶ Horn Point Laboratory, University of Maryland Center for Environmental Science, Cambridge, MD, United States ¶ Corresponding author * These authors contributed equally.

DOI: [10.xxxxxx/draft](https://doi.org/10.xxxxxx/draft)

Software

- [Review](#)
- [Repository](#)
- [Archive](#)

Editor: [✉](#)

Submitted: 26 March 2026

Published: unpublished

License

Authors of papers retain copyright and release the work under a Creative Commons Attribution 4.0 International License ([CC BY 4.0](#)).

Summary

Small aerial drones, or unoccupied aerial systems (UAS), conveniently achieve scales of observation between satellite resolutions and in situ sampling, effectively diminishing the “blind spot” between these established measurement techniques (P. C. Gray et al., 2022). UAS equipped with off-the-shelf multispectral sensors originally designed for terrestrial applications are being increasingly used to derive water quality properties. Multispectral UAS imagery requires post processing to radiometrically calibrate raw pixel values to useful radiometric units such as reflectance. In aquatic applications, there are additional steps to remove surface reflected light and sun glint, and different approaches to estimate water quality parameters. Georeferencing and mapping UAS imagery over water also comes with challenges since typical structure from motion photogrammetry techniques fail due to lack of feature matching. DroneWQ can 1) convert raw multispectral imagery to total radiance (Lt) with units of W m⁻² nm⁻¹ sr⁻¹, 2) remove surface reflected light (Lsr) to calculate water leaving radiance (Lw), 3) measure downwelling irradiance (Ed) from either the calibrated reflectance panel, downwelling light sensor (DLS), or a combination of the two, 4) calculate remote sensing reflectance (Rrs) by dividing Ed by Lw, and 5) mask pixels containing specular sun glint or instances of vegetation, shadowing, etc., 6) use Rrs as input into various bio-optical algorithms to derive chlorophyll a and total suspended sediment concentrations, and 7) georeference using image metadata and sensor specifications to orient and map to a known coordinate system.

Statement of need

DroneWQ is a Python package for multispectral UAS imagery processing to obtain remote sensing reflectance (Rrs), the fundamental input into ocean color algorithms which can be used to estimate and map water quality parameters. The processing steps, calibrations, and corrections necessary to obtain research quality Rrs data from UAS can be prohibitively difficult for those who do not specialize in optics and remote sensing, yet this data can reveal entirely new insight into aquatic ecosystems. DroneWQ was designed to be a simple pipeline for those who wish to utilize UAS multispectral remote sensing to analyze ocean color and water quality. The simple functionality of DroneWQ will enable effective water quality monitoring at fine

spatial resolutions, leading to exciting scientific exploration of UAS remote sensing by students, scientists, and water quality managers.

State of the field

Private software and applications exist for UAS post-processing, georeferencing, and mapping (e.g., Pix4D, Agisoft Metashape); however, these tools are primarily designed for terrestrial imagery and often perform poorly over water due to lack of stable features for image matching. The MicaSense company (now EagleNXT) developed a codebase for Python-based image processing of MicaSense RedEdge and Altum imagery (<https://github.com/micasense/imageprocessing>) which is suitable for basic terrestrial remote sensing.

DroneWQ builds upon these workflows to enable the calculation of aquatic remote sensing reflectance and derived water quality parameters. To our knowledge, DroneWQ is the first open-source software package that allows users to process raw multispectral UAS imagery into aquatic remote sensing reflectances and water quality metrics relevant to environmental research and management.

Software design

DroneWQ is a modular, object oriented Python package that implements a processing pipeline for transforming raw multispectral UAS imagery into georeferenced remote sensing reflectance and derived water quality products. The software follows a workflow-oriented design in which individual processing steps (data ingestion, radiometric calibration, reflectance retrieval, quality control, bio-optical inversion, georeferencing, and mosaicking) are implemented in reusable classes. This architecture promotes transparency, flexibility, and ease of extension, allowing users to modify or replace individual components without altering the overall pipeline. A key design feature of DroneWQ is the ability to project and georeference imagery, which leverages onboard GPS and orientation metadata to map imagery without relying on conventional structure-from-motion approaches that often fail over water due to limited spatial feature contrast. The package integrates radiometric corrections, glint correction methods, and established ocean color algorithms to produce remote sensing reflectance and water quality products within a unified framework. Configuration is handled through a centralized setup function that manages file paths and processing parameters, supporting reproducible workflows. The software has fairly minimal dependencies and can be integrated into a wide range of development workflows. Here it is demonstrated within a Jupyter notebook, enabling interactive analysis while maintaining reproducibility for batch processing.

Background/Theory

UAS can measure remote sensing reflectance (Rrs) defined as:

$$\text{Eq. 1} \quad R_{rs}(\theta, \varphi, \lambda) = LW(\theta, \varphi, \lambda) / Ed(\theta, \varphi, \lambda)$$

where LW ($\text{W m}^{-2} \text{ nm}^{-1} \text{ sr}^{-1}$) is water-leaving radiance, Ed ($\text{W m}^{-2} \text{ nm}^{-1}$) is downwelling irradiance, θ represents the sensor viewing angle between the sun and the vertical (zenith), φ represents the angular direction relative to the sun (azimuth) and λ represents wavelength.

UAS do not measure Rrs directly as the at-sensor total radiance (LT, $\text{W m}^{-2} \text{ nm}^{-1} \text{ sr}^{-1}$) constitutes the sum of LW and incident radiance reflected off the sea surface into the detector's field of view, referred to as surface reflected radiance (LSR). LW is the radiance that emanates from the water and contains a spectral shape and magnitude governed by optically active water constituents, while LSR is independent of water constituents and instead governed by

the water surface reflecting the downwelling light; a familiar example is sun glint. Here we define UAS total reflectance (R_{UAS}) as:

$$\text{Eq. 2} \quad R_{UAS}(\theta, \phi, \lambda) = LT(\theta, \phi, \lambda) / E_d(\lambda)$$

where

$$\text{Eq. 3} \quad LT(\theta, \phi, \lambda) = LW(\theta, \phi, \lambda) + LSR(\theta, \phi, \lambda)$$

Due to the differing orientation of wave facets reflecting radiance from different parts of the sky, LSR can vary widely within a single UAS image. 'DroneWQ' provides multiple options from the literature for removing LSR.

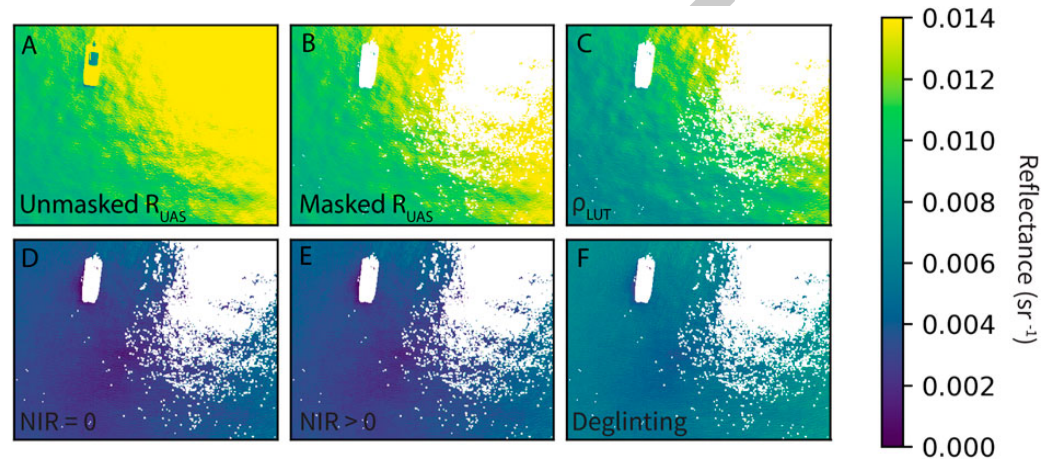


Figure 1. Example of an individual UAS image (green band) at different processing steps and methods: (A) R_{UAS} , (B) R_{UAS} with initial sun glint masking and (C–F) remote sensing reflectance (R_{rs}) using various methods to remove surface reflected light: (C) look-up table (LUT) from HydroLight simulations, (D) Dark pixel assumption with $NIR = 0$, (E) Dark pixel assumption with $NIR > 0$, (F) Deglinting methods following (Hedley et al., 2005). Figure taken from (Windle & Silsbe, 2021).

A secondary challenge with aquatic UAS remote sensing is georeferencing and mosaicking imagery. Current photogrammetry techniques (e.g. Structure-from-Motion (SfM)) are not capable of stitching UAS images captured over large bodies of water due to a lack of key points in images of homogenous water surfaces. DroneWQ uses sensor pose information to project and mosaick imagery.

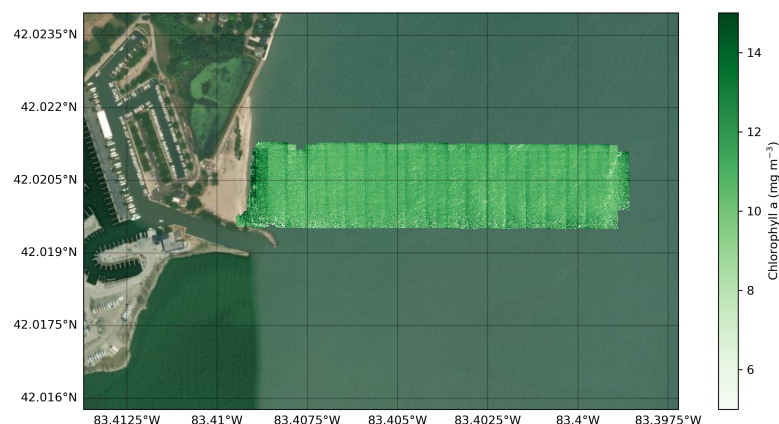


Figure 2. Final orthomosaic of UAS images collected over Western Lake Erie processed to

108 chlorophyll a concentration.

109 Research impact statement

110 DroneWQ has demonstrated growing research impact and community engagement since its
 111 initial development. The project has expanded beyond its original authors ((?); (?)) to include
 112 contributions from four additional developers, reflecting increasing adoption and collaborative
 113 development. Community involvement has supported the evolution of the software through
 114 feature requests, bug reports, and user feedback submitted via GitHub Issues and direct
 115 communication, indicating active use by a broader research community. DroneWQ has been used
 116 in multiple peer-reviewed studies focused on UAS aquatic remote sensing and water quality
 117 retrievals (W. Gray P. C., 2022; Windle & Silsbe, 2021; ?). These applications demonstrate
 118 the utility of the software for advancing research using multispectral UAS imagery in aquatic
 119 environments, particularly where traditional processing approaches are limited. As interest in
 120 UAS-based water quality monitoring continues to grow, DroneWQ provides an open, extensible
 121 framework that supports both scientific investigation and applied environmental monitoring.

122 AI usage disclosure

123 The creation of this software was mostly done by humans. AI assisted in identifying and
 124 diagnosing bugs, as well as generating docstrings, both of which were thoroughly reviewed by
 125 humans.

126 Acknowledgements

127 We acknowledge and appreciate helpful support from the Micasense team. We thank Julian
 128 Dale for assisting with UAS flights.

129 References

- 130 Gray, P. C., Larsen, G. D., & Johnston, D. W. (2022). Drones address an observational
 131 blind spot for biological oceanography. *Frontiers in Ecology and the Environment*, 20(7),
 132 413–421. <https://doi.org/10.1002/fee.2472>
- 133 Gray, W., P. C. (2022). Robust ocean color from drones: Viewing geometry, sky reflection
 134 removal, uncertainty analysis, and a survey of the gulf stream front. *Limnology and*
 135 *Oceanography: Methods*, 20(10), 656–673. <https://doi.org/10.1002/lom3.10511>
- 136 Hedley, J. D., Harborne, A. R., & Mumby, P. J. (2005). Technical note: Simple and robust
 137 removal of sun glint for mapping shallow-water benthos. *International Journal of Remote*
 138 *Sensing*, 26(10), 2107–2112. <https://doi.org/10.1080/01431160500034086>
- 139 Windle, A. E., & Silsbe, G. M. (2021). Evaluation of unoccupied aircraft system (UAS) remote
 140 sensing reflectance retrievals for water quality monitoring in coastal waters. *Frontiers in*
 141 *Environmental Science*, 9. <https://doi.org/10.3389/fenvs.2021.674247>